Write a program to generate Association Rules using the Apriori algorithm.

Topics you need to learn

1. Association Rule Mining

It discovers relationships between items in a dataset.

Example: Customers who buy bread may also buy butter.

$$\text{Bread} \rightarrow \text{Butter}$$

2. Transaction Dataset

Data is stored as transactions containing items.

Transactin	Items
T1	Bread, Milk
T2	Bread, Butter
T3	Bread, Milk, Butter
T4	Milk, Butter

Each row represents one transaction.

3. Support

Support measures how frequently an item or item combination occurs in all transactions.

For a rule A:

$$Support(A) = \frac{Transactionscontaining A}{Totaltransactions}$$

If Bread appears in 3 of 4 transactions:

$$Support(Bread) = \frac{3}{4} = 75\%$$

For a rule $A \rightarrow B$:

$$\boxed{Support(A \rightarrow B) = \frac{Count(A \cap B)}{N}}$$

- A = antecedent (left side)
- B = consequent (right side)
- N = total number of transactions
- $Count(A \cap B)$ = transactions containing both A and B

4. Apriori Algorithm

Apriori finds frequent item combinations by gradually increasing their size.

- First: individual items
- Next: pairs of items
- Then: groups of three items

Key rule: If an itemset is infrequent, its larger supersets cannot be frequent.

5. Confidence

Confidence measures how often the consequent appears when the antecedent appears.

$$Confidence(A \rightarrow B) = \frac{Support(A \cup B)}{Support(A)}$$

For example, if Bread appears 3 times and Bread with Milk appears 2 times:

$$Confidence(Bread \rightarrow Milk) = \frac{2}{3} = 66.67\%$$

6. Lift

Lift helps determine whether two items occur together more often than expected from their individual frequencies.

$$Lift(A \rightarrow B) = \frac{Confidence(A \rightarrow B)}{Support(B)}$$

- Lift greater than 1: positive association
- Lift equal to 1: no association under this measure
- Lift less than 1: negative association

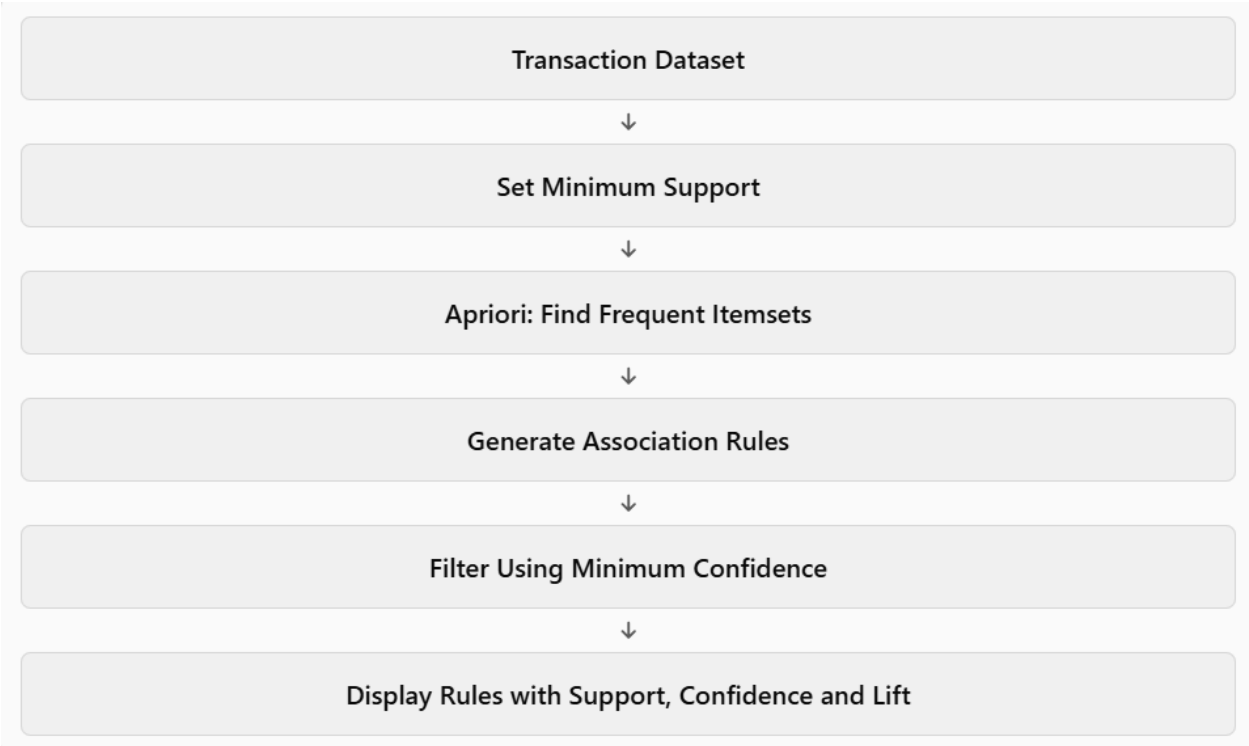
Positive association means two items tend to occur together.

Example: Customers who buy bread often also buy butter.

In the Apriori algorithm, if the lift is greater than 1, it indicates a positive association between bread and butter.

In simple words: **Buying one item is associated with a higher chance of buying the other item**.

The complete process



Process of code explanation

1. Input features

Your original dataset has numerical features:

Wine	alcohol	color_intensity	proline
W1	12.5	5.2	800
W2	13.8	3.0	1200
W3	11.9	6.5	600

These columns are called **features**.

2. After categorization

Your code converts numerical values into categories such as Low and High.

Wine	alcohol	color_intensity	proline
W1	Low	High	Low
W2	High	Low	High
W3	Low	High	Low

3. After one-hot encoding

The categories become individual items.

alcohol_Low	alcohol_High	color_intensity_Low	color_intensity_High	proline_Low	proline_High
True	False	False	True	True	False
False	True	True	False	False	True
True	False	False	True	True	False

Now Apriori treats each row as a transaction.

How many possible itemsets?

For n distinct items, the number of possible non-empty itemsets is:

$$\boxed{2^n - 1}$$

Since your dataset has 8 items:

$$2^8 - 1 = 256 - 1 = \boxed{255}$$

So there are **255 possible non-empty itemsets** before applying the support threshold.

Breakdown by itemset size

Itemset size	Calculation	Possible combinations
1 item	$\binom{8}{1}$	8
2 items	$\binom{8}{2}$	28
3 items	$\binom{8}{3}$	56
4 items	$\binom{8}{4}$	70
5 items	$\binom{8}{5}$	56
6 items	$\binom{8}{6}$	28
7 items	$\binom{8}{7}$	8
8 items	$\binom{8}{8}$	1
Total		**255**

1. Take a transaction dataset

Suppose we have these 5 transactions after one-hot encoding.

Transactin	Milk	Bread	Butter
T1	1	1	0

T2	1	1	1
T3	0	1	1
T4	1	1	1
T5	0	0	1

Here, `1` means present and `0` means absent.

2. Step 1: Generate one-item itemsets

First, Apriori considers each item separately.

Itemset	Count	Support
`{Milk}`	3	$3/5 = 60\%$
`{Bread}`	4	$4/5 = 80\%$
`{Butter}`	4	$4/5 = 80\%$

Suppose `min_support = 0.4`, or 40%.

All three items are frequent because their support is at least 40%.

3. Step 2: Generate two-item itemsets

Apriori combines the frequent individual items into pairs.

Example A: `{Milk, Bread}`

Milk and Bread must both be present in the same transaction.

Transaction	Milk	Bread	Both present?
T1	1	1	Yes
T2	1	1	Yes
T3	0	1	No
T4	1	1	Yes
T5	0	0	No

Count = 3.

$$Support(Milk, Bread) = \frac{3}{5} = 60\%$$

Therefore, `{Milk, Bread}` is frequent.

Example B: `{Milk, Butter}`

Both items are present in T2 and T4.

$$Support(Milk, Butter) = \frac{2}{5} = 40\%$$

This itemset is also frequent.

Example C: `{Bread, Butter}`

Both items are present in T2, T3 and T4.

$$Support(Bread, Butter) = \frac{3}{5} = 60\%$$

This itemset is frequent too.

4. Step 3: Generate a three-item itemset

Now combine all three items:

`{Milk, Bread, Butter}`

All three must occur together in the same transaction.

Transaction	Milk	Bread	Butter	All three present?
T1	1	1	0	No
T2	1	1	1	Yes
T3	0	1	1	No
T4	1	1	1	Yes
T5	0	0	1	No

Count = 2.

$$Support(Milk, Bread, Butter) = \frac{2}{5} = 40\%$$

Since support equals the minimum support of 40%, this itemset is frequent.

Calculate the possible rules from your 28 itemsets

The formula is:

$$\text{Possible rules} = 2^k - 2$$

Here, k is the number of items in a particular itemset.

2. Calculate the possible rules from your 28 itemsets

Your 28 itemsets consist of:

Itemset size	Number of itemsets	Rules per itemset	Total possible rules
1 item	8	0	0
2 items	16	2	32
3 items	4	6	24
Total	**28**		**56**

Therefore, assuming your 28 itemsets have this size distribution, they can generate up to:

$$32 + 24 = \boxed{56 \text{ possible rules}}$$

The actual number may be lower if the algorithm does not generate every candidate or if rules fail the confidence threshold.

3. How does the function filter those rules?

Suppose the generated rules have these confidence values:

Rule	Confidence	Decision
A → B	85%	Keep
B → A	60%	Remove
A, B → C	75%	Keep
A, C → B	65%	Remove

Your setting is: min_threshold=0.7

How are rules generated from an itemset containing 3 items?

Suppose your frequent itemset is:

$$\{A, B, C\}$$

Here, A, B, and C could represent `alcohol_Low`, `color_intensity_Low`, and `proline_Low`.

The possible rules are:

No.	Antecedent (left side)	Consequent (right side)
1	A	B, C
2	B	A, C
3	C	A, B
4	A, B	C
5	A, C	B
6	B, C	A

For three items, there are:

$$2^3 - 2 = \boxed{6 \text{ possible rules}}$$

The formula excludes the empty set and the full set as possible sides of a rule.

4. How is confidence calculated?

Consider the rule:

$$\{A, B\} \rightarrow \{C\}$$

The formula is:

$$Confidence(A, B \rightarrow C) = \frac{Support(A,B,C)}{Support(A,B)}$$

Suppose the Wine dataset produces these counts:

- Total wine samples = 178
- Samples containing both A and B = 70
- Samples containing A, B and C together = 53

First, calculate the support values:

$$Support(A, B) = \frac{70}{178} = 0.3933$$

$$Support(A, B, C) = \frac{53}{178} = 0.2978$$

Now calculate confidence:

$$Confidence = \frac{0.2978}{0.3933}$$

Equivalently, using the counts:

$$Confidence = \frac{53}{70} = 0.7571$$

$$\boxed{Confidence = 75.71\%}$$

Since 75.71% is greater than 70%, the rule passes the confidence threshold.